

Keep the Angle, in Practice: A Substrate-Agnostic Manifold Diagnostic and Its Failure Modes

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Abstract

The companion theory paper (*Keep the Angle*, Paper I) proves that the angular coordinate of a low-Laplacian spectral embedding preserves geodesic geometry while the radial coordinate is degree/density nuisance. This paper is the empirical study of that basis as a *practical, substrate-agnostic diagnostic*. On graph families spanning random-geometric tori and king-lattices, manifold-embedding samples (swiss-roll), Lorentzian causal sets, and hypergraph-rewriting (Wolfram-model) substrates, the angular map recovers graph geodesics wherever the substrate carries genuine low-dimensional Riemannian geometry, independent of how the graph was generated: on a 2-torus the angular Spearman correlation is ≈ 0.92 and flat in the mode count, while the full commute embedding decays; in a bake-off it beats every closed-form spectral competitor (0.894 ± 0.008 vs. $0.60\text{--}0.77$ on the 2-torus over five independent draws). We show the same map is a *diagnostic*: a two-factor small-world/dimension screen separates genuine manifolds from destroyed or non-manifold geometry, angular fidelity declines monotonically with measured intrinsic dimension across five Wolfram-model rule families (an exploratory trend, $\text{angle-}\rho \approx 1.09 - 0.157d$), and under continuum refinement the basis *sharpens* on sampled manifolds while *decaying* on emergent hypergraph geometry—returning opposite verdicts for sampled and emergent substrates. Two pre-registered physical predictions are honest negatives. Everything here is pre-registered and reproducible; the theory it rests on is Paper I.

1 Introduction

Keep the Angle (Paper I) establishes, as theorems, that row-normalizing a low-mode spectral embedding—projecting each node to the unit sphere and keeping only its direction—recovers the graph’s geodesic geometry, because the discarded radial coordinate is degree/density nuisance (von Luxburg-degenerate for the full unnormalized commute radius at $d \geq 3$) while the angular coordinate carries the metric. That paper is about *why* the angle works: an exact normalization identity, a deterministic angular-transfer theorem, a conditional graph-to-continuum result, the unconditional flat torus and sphere instances, a uniform lower angular bound on flat tori, and the filter dichotomy (heat uniform, commute one-sided, plain rank-collapsing).

This paper is about *whether, where, and how well* the angle works in practice, and what it does when the geometry is not there. Our thesis is that the angular basis is a *substrate-agnostic manifold diagnostic*: on any graph that carries genuine low-dimensional Riemannian geometry it recovers that geometry, independent of graph origin; on graphs that do not, its failure is *predictable and informative* rather than a silent artifact. We support this with a controlled measurement program.

Contributions. (1) A geometry-recovery benchmark across five substrate families with matched controls (§3), and a bake-off in which the angle beats every purely spectral distance built from the

same spectrum (§3.1). (2) A study of *emergent* geometry from hypergraph rewriting—Wolfram-model substrates that are geometry with no latent coordinates—including an exploratory fidelity-versus-dimension trend and temporal stability across substrate evolution (§4). (3) The angular basis as an explicit *diagnostic*: a two-factor screen that separates manifolds from non-manifolds, and a continuum-refinement test that dissociates sampled from emergent geometry (§5). (4) Two honest negatives (§6) and a scoped physical interpretation (§7). We do not re-prove Paper I; theorem references below point there.

2 The angular basis and measurement protocol

Given a graph with normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k) , the low-mode embedding places node i at $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$, with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/r_i$. We score the angular geometry by the Spearman rank correlation (“angle- ρ ”) between the chordal angular distances $\|\hat{u}_i - \hat{u}_j\|$ and true graph geodesics over a fixed anchor set, against three controls: *random* (m random modes), *magnitude* (radius only), and *Isomap* (classical MDS of the geodesics, an oracle given the geodesic matrix). Intrinsic dimension d is measured by three estimators (ball-growth, spectral dimension, effective rank); low inter-estimator spread certifies a manifold. Protocol details, draw counts, and reproducibility tiers mirror Paper I; all runs are pre-registered (PREREG.RUNG3/4) and released in the repository, <https://github.com/ahb-sjsu/the-angular-observer>.

3 Geometry recovery across substrates

On a 2-torus ($n = 2000$) the polar decomposition separates cleanly: the angle carries the geometry (flat across the gap-closed cutoffs 4, 8, 12, 20; e.g. the $m=20$ value is 0.891 ± 0.009 over five independent draws), the magnitude is uninformative, and the full commute embedding is worse and decays with m . A random-mode basis reconstructs geodesics at $\rho \approx 0$.

Across substrate families (Table 1), every graph with genuine low- d Riemannian geometry shows the effect, independent of origin; the metric fails exactly where the substrate fails the manifold criterion—the Lorentzian causal set (whose dimension estimators scatter over 2.9–10.7) and the rewired lattice—the predicted failures, not exceptions. Two caveats keep the claim honest. On the two flat substrates classical MDS beats the angle (it wins where the manifold embeds near-isometrically in low-dimensional Euclidean space); the angle wins where topology obstructs such an embedding (the torus) and, unlike MDS, is uniform in m and n . And the swiss-roll is a manifold-embedding case: the basis is identical whether the manifold comes from hypergraph rewriting or from points sampled on a continuous manifold.

Table 1: Geometry recovery across substrate families ($m=20$, 150 anchors).

substrate	meas. d	angle- ρ	random	Isomap $_{d=2}$
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll (embedding)	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	2.9–10.7	0.57	0.01	0.55
small-world (destroyed)	3.6–4.3	0.45	0.00	—

3.1 A bake-off against named spectral distances

Scoring every closed-form spectral distance on the identical graph and anchor set (Table 2), the angle beats each purely spectral competitor, and the margin widens in $d=3$ where commute degeneracy bites harder. The most informative row is the amplified commute distance of von Luxburg, Radl and Hein, designed precisely to undo commute degeneracy by *subtracting* the $1/k_i + 1/k_j$ term: it barely improves on raw commute and stays far below the angle, because subtracting the degenerate radial term is not the same as *dividing it out*, which is what row-normalization does (Paper I). Isomap wins only because it is handed the geodesics—an oracle ceiling, not a competitor; the angle recovers 0.83–0.89 of it from the spectrum alone. Angle scores replicate across five independent graph draws: 0.894 ± 0.008 (2-torus, min–max 0.886–0.907) and 0.831 ± 0.007 (3-torus, 0.818–0.836).

Table 2: Bake-off: Spearman ρ vs. graph geodesics; angle scores are five-draw means.

distance	2-torus	3-torus
angle (ours)	0.894	0.831
full commute / resistance	0.601	0.360
biharmonic	0.735	0.515
diffusion distance (best t)	0.771	0.631
amplified commute	0.628	0.364
<i>Isomap, given the geodesics</i>	<i>0.971</i>	<i>0.952</i>

3.2 Which weighting carries the angle, and density normalization

The direction depends on the per-mode weight. Plain eigenmaps ($w_k=1$) give an angle that *collapses* with mode count (angle- ρ falls from 0.554 ± 0.013 at $m=10$ to 0.081 ± 0.012 at $m=20$ on the torus, five draws), because equal weighting lets high modes swamp the direction; the commute weight $1/\sqrt{\lambda_k}$ is flat instead ($0.918 \pm 0.006 \rightarrow 0.896 \pm 0.006$), as is diffusion. Degree correction is *invisible* to the angle: dividing or multiplying each row by $\sqrt{k_i}$ leaves angle- ρ unchanged to three digits, confirming the degree lives in the radius, not the angle (Paper I). Under strongly non-uniform sampling the raw angle is *not* sampling-invariant (angle- ρ degrades from 0.93 to 0.79 as a density contrast grows to $45\times$), because the graph Laplacian converges to a density-weighted operator; the full Coifman–Lafon $\alpha=1$ normalization restores it exactly (to 0.92–0.96, flat across the contrast range), isolating density into the radius and geometry into the angle.

4 Emergent geometry from hypergraph rewriting

Wolfram-model substrates are geometry with *no latent coordinates*: a manifold, if it appears, emerges from hypergraph rewriting rather than from sampling a continuous space. Random arity-2 rules yield clean but only ≈ 1.5 D manifolds; clean 2D is rare but not absent, reached both by a borrowed rule ($d = 2.07 \pm 0.01$) and by from-scratch evolutionary search. The angular basis holds on genuine 2D emergent space (Table 1).

An exploratory dimension–fidelity trend. With only five independent rule families, no “law” can responsibly be claimed. Across 20 manifolds (5 rules $\times$ 4 correlated seeds, $d \approx 1\text{--}3.6$), angular fidelity falls with measured intrinsic dimension, angle- $\rho \approx 1.09 - 0.157d$, with a rule-clustered

descriptive interval $[-0.343, -0.094]$ four times wider than a naive point bootstrap; we read this as a five-family exploratory monotonic trend, not an inferential interval. The mechanism is the dimension-gated degeneracy of Paper I; its rank-specific bound is consistent with, but does not derive, the observed decay.

Stability across substrate evolution. Robustness across substrate *types* is a statement about fixed graphs; the basis also persists as a substrate *evolves*. Embedding successive rewrite snapshots of 151 qualifying rules on a fixed anchor set, the angular structure tracks the evolving geometry at least as stably as the graph’s own geodesics on every rule resolvable at protocol scale (143/143 against a pre-registered 80% bar), and over the longest time lever angular persistence declines only gently with emergent dimension ($-0.029/\text{dim}$), replicating independently in both arities.

5 The angular basis as a diagnostic

A two-factor screen. A small-world coefficient ω and the inter-estimator dimension spread separate genuine manifolds ($\omega \leq -0.50$, spread ≤ 0.51) from strongly destroyed or non-manifold geometry ($\omega \geq +0.23$, spread ≥ 0.77). The low angle- ρ on the rewired lattice and the Lorentzian causal set is thus small-world/non-manifold contamination, the predicted failure of the basis on non-geometry, not a failure on genuine geometry. A king-lattice rewiring sweep bounds the screen honestly: angle- ρ collapses almost at once ($0.81 \rightarrow 0.47$ by 1% rewiring) while ω does not cross zero until $\approx 5\%$, so angular fidelity is itself the finer detector of geometric corruption in the transition band.

Continuum survival, and a dissociation. Refining each substrate toward the continuum with a sparse eigensolver ($N \in \{2, 5, 10, 20, 50\} \times 10^3$) dissociates the two substrate families. On random geometric graphs the angular fidelity is flat-to-rising in N (RGG $d=3$: $0.856 \rightarrow 0.912$, Spearman($\rho, \log N$) = $+1.00$): refining the mesh makes the low modes resolve the geometry *better*. The emergent R_{3D} manifold instead *decays* monotonically ($0.749 \rightarrow 0.654$, Spearman -1.00), falling below a pre-registered floor. A sampled manifold refines to a clean continuum; an emergent hypergraph, under this basis, does not. The angular basis is thus a diagnostic for whether a substrate’s fine structure stays manifold-like under refinement, returning different verdicts for sampled (2/3 pass) and emergent (1/1 fail) geometry. One qualification: the dissociation is stated at a fixed mode budget; a spectral-gap-matched rerun inverts the direction, so “clean geometry” is a relation between mode-budget resolution and substrate scale—the motivation for a dimension-adaptive basis.

6 Honest negatives

Two pre-registered predictions failed. **Not secretly hyperbolic:** against a binary-tree positive control, the tested substrates show no signature of latent hyperbolic geometry. **No dynamical, energy-responsive curvature:** the strongest testable claim of the observer-relative reading—that the coarse-grained curvature responds dynamically to injected energy—is a null. We report both; neither undermines the basis-level and diagnostic results above, and the physical interpretation is offered only as motivation.

7 Physical interpretation and outlook

The decomposition “separate scale from direction, keep the part that carries the task geometry” recurs beyond graph geometry: polar-coordinate KV-cache quantization splits key vectors the same way, and the recurrence is evidence for the *decomposition*, not for always discarding the radius—the boundary being exactly whether the discarded radial variable is nuisance for the metric preserved (the tangential-noncollapse condition of Paper I). For attention keys, whose geometry depends on per-channel scale, discarding the radius destroys generation: scale is signal there, not nuisance. In the observer-relative reading of emergent-geometry programs, the coarse-graining a bounded observer performs to perceive geometry is a candidate for exactly this angular projection, but its strongest dynamical claim is the null of §6, so we leave that door open. The operational contribution stands independently: the angular basis is a substrate-agnostic diagnostic for low-dimensional Riemannian structure, with predictable, informative failure modes. Open directions: a dimension-adaptive basis that tunes mode-count and diffusion time against the fidelity decay; substantially more independently selected rule families to test the dimension trend; and evaluation against the latent manifold metric wherever coordinates exist.

Reproducibility. All pre-registration documents, experiment scripts, per-run result files, rule sets, and random seeds are released in the accompanying repository, <https://github.com/ahb-sjsu/the-angular-observer>. The theory these experiments test is Paper I (*Keep the Angle*).

References

- [1] A. H. Bond. *Keep the Angle: A Geometry-Preserving Basis in Spectral Embeddings* (Paper I). Manuscript, 2026.